The Components of Content

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1 Introduction

Is the content of a belief in the head of the believer, in the environment, or in a combination of the two? Certainly not in the head, according to recent thought. Persuasive thought-experiments suggest that two physically identical believers might have different belief contents, due to differences in their environment. Where I believe that water is wet, my twin on Twin Earth[1] —which is just like Earth, except that H_2O is replaced by the superficially identical XYZ— believes something quite different. His belief is not about water but about twin water: I believe that water is wet, but he believes that twin water is wet. Similarly, I might falsely believe that I have arthritis in my thigh, but my twin in a different linguistic community—where "arthritis" is used to refer to an inflammation of both the joints and the muscles—has a different belief, not about arthritis but about a disease that we may call twarthritis. Where I believe that I have arthritis in my thigh, my twin believes that he has twarthritis in his thigh. So the content of a belief is not determined by the internal state of the believer.

Still, there remains a strong intuition that there is some kind of content that my twins and I share. Even though we refer to different things, there seems to be a sense in which we both take the world to be the same. Certainly our physical behavior is the same, so that insofar as content causes and explains behavior, it is plausible to suppose that internal aspects are doing the work. Intuitions such as these have led some to postulate a special kind of content—*narrow* content—which is determined by the internal state of the believer.[2] Every belief has both narrow content and wide content. Twins' beliefs may differ in their wide content, but their narrow content is always the same.

[1] For Twin Earth, see H. Putnam, "The Meaning of 'Meaning"', in *Language, Mind, and Knowledge*, ed. K. Gunderson (Minneapolis: University of Minnesota Press, 1975). For arthritis, see T. Burge, "Individualism and the Mental", *Midwest Studies in Psychology* 4 (1979):73-122.

[2] Arguments for narrow content can be found in J. Fodor, *Psychosemantics* (Cambridge, MA: MIT Press, 1987); D. Lewis, "Reduction of Mind", in *Companion to the Philosophy of Mind*, ed. S. Guttenplan (Oxford: Basil Blackwell, 1994); B. Loar, "Social Content and Psychological Content", in *Contents of Thought*, ed. R.H. Grimm and D.D. Merrill (Tucson: University of Arizona Press, 1988); S. White, "Partial Character and the Language of Thought", *Pacific Philosophical Quarterly* 63 (1982):347-65.

The road from intuition to theory has been a difficult one, and no concrete account of narrow content has yet gained wide acceptance. Indeed, narrow content is thought by many to face insuperable difficulties. Perhaps the greatest difficulty is that because narrow content is purely internal, it seems to lack an appropriate relation to the external world, the sort of relation that is surely required to qualify as *content*. Such external notions as truth and reference are thought to be in the domain of wide content. Even those who have advocated narrow content have often conceded that it is not the sort of thing that carries *truth-conditions*.[3] The twin cases show us that truth-conditions are not internally determined, so that only wide content is the sort of thing that can be true and false. Supporters of narrow content have argued that this is not a problem, and that narrow content is tied to a different sort of semantics, such as that of an internal conceptual role. Critics have argued that this shows that narrow content cannot be *content* at all.[4]

I will argue that this line of thinking is entirely misguided. There is a robust and natural notion of narrow content such that narrow content has truth-conditions of its own. Beliefs not only have two sorts of content; they also have two sorts of truth-conditions, narrow and wide. This can be seen by developing the insight, implicit in much contemporary work in the philosophy of language, that content is two-dimensional. In this way we can arrive at a truly semantic notion of narrow content.

It turns out that the content of any thought decomposes naturally into two components, each with associated truth-conditions: its *notional* and *relational* content. Notional content is always narrow, but relational content is frequently wide. I will argue that notional content determines the cognitive and rational relations between thoughts, and that notional content governs our actions. Further, the decomposition strategy shows us why the "wideness" of ordinary belief ascriptions is compatible with the primacy of narrow content in the governance of action. Ordinary belief ascriptions ascribe a combination of notional and relational content; it is in virtue of the relational component that the ascribed content is wide, and in virtue of the notional component that the ascribed content is explanatory. By recognizing the two-dimensional nature of content, many puzzles in the philosophy of mind and language can be dissolved.

2 First and Second Intensions[5]

[3]Fodor, *Psychosemantics*, p. 50; N. Block, "Functional Role and Truth-Conditions", *Proceedings of the Aristotelian Society* 61 (1988):157-81.

[4]Adams, F., Drebushenko, D, Fuller, G., and Stecker, R., "Narrow Content: Fodor's Folly", *Mind and Language* 5 (1990): 213-29.

[5]The two-dimensional account of the contents of thought developed in what follows bears a formal and conceptual resemblance to the accounts of the semantics and pragmatics of language in D. Kaplan, "Demonstratives", in *Themes from Kaplan*, ed. J. Almog, J. Perry, and H. Wettstein (Oxford: Oxford University Press, 1989), and R. Stalnaker, "Assertion", in *Syntax and Semantics: Pragmatics, Vol. 9*, ed. P. Cole (Academic Press, 1978), although it was more directly inspired by S.A. Kripke's *Naming and Necessity* (Cambridge, MA: Harvard University Press, 1980), and Putnam's "The Meaning of 'Meaning"'. There is also an obvious debt to work cited below by Davies

In what follows, an *intension* is a function over the space of possible worlds, taking a world as argument and returning an entity in that world. A *centered world* is a possible world with a marked individual and time (the "center" of that world).[6] W and W^* are the spaces of possible worlds and centered worlds respectively, and R is the space of entities in those worlds. A *concept*, pre-theoretically, is an element of a specific thought, of the sort that is an appropriate candidate for reference—that is, it either refers or has empty reference. (Note that a concept is a token rather than a type, tied to a specific thinker on a specific occasion.) A concept, in thought, is the analog of a term in language. For ease of exposition I will restrict attention to concepts that are associated with terms in the thinker's language ("I", "water", "arthritis"), but the account can easily be extended to concepts without linguistic counterparts.

It is a familiar idea that every concept has an associated intension, mapping a possible world to the referent of the concept within that world. It is less commonly acknowledged that there are in fact two intensions associated with a concept, although this point is implicit in much recent work on meaning and reference. It is central to the work of Kripke and Putnam, for example, that there are two ways in which the reference of a concept in a world can depend on that world. First there is a dependency whereby reference is fixed in the actual world, depending on how the world turns out. My concept "water" refers to H_2O given that the world has turned out one way, but if it had turned out differently my concept might have referred to XYZ. Second, there is a dependency whereby reference in counterfactual worlds is determined, given that the actual world is held constant. For example, given that the actual world turns out as it has, my concept "water" picks out H_2O in all counterfactual worlds.

These dependencies can be formalized as follows. The *first intension* is a function $f_1 : W^* \rightarrow R$ from centered worlds to referents, returning what the concept would refer to if that centered world were actual. This expresses the way that the reference of the concept is fixed, depending on how the actual world turns out—that is, depending on which centered world turns out to be the actual world of the thinker. The *second intension* is a function $f_2 : W \rightarrow R$ from uncentered worlds to referents, returning the referent of the concept in a counterfactual world, given that the actual world in which the concept is deployed has already been fixed. For example, the first intension of my concept "I" is the function that maps a centered world to the individual at its center; the second intension is the function that picks out David Chalmers at every world in which he exists.

To evaluate a concept's first intension at a given centered world, we must *consider that world as actual*, asking: to what would the concept refer, if the actual world turned out this way?[7] For example, if the actual world turned out to contain XYZ as the dominant clear liquid in the oceans and lakes on my planet, then my concept "water" would refer to XYZ. So the first intension maps the XYZ-world to XYZ. Note that an uncentered world will not suffice as

and Humberstone and Evans on two-dimensional modal logic, and by Lewis and Perry on the content of thoughts.

[6]This notion is introduced in W.V. Quine's "Propositional Objects", *Critica* 2:5 (1968):3-22, where a centered world is defined as a world with a marked space-time point. The definition above is due to D. Lewis, "Attitudes *De Dicto* and *De Se*", *Philosophical Review* 88 (1979): 513-43.

an actual-world candidate here: concepts such as "I" and "water" indicate that there is often an element of indexicality in the fixation of reference. If it is specified that the actual world contains both H_2O and XYZ (on otherwise similar planets, say), the reference of my concept "water" is undetermined until we specify where in that world *I* am located. In general, an actual-world candidate for our purposes will be a world-as-I-find it, marked with a viewpoint at its center.

It is an interesting fact about concepts that for a very wide range of centered worlds, the reference of the concept in that world (considered as actual) is determinate and accessible to the thinker. To see this, note that for such a world, if that world *did* turn out to be actual—even if this were a very surprising development—then once we knew all about the qualitative structure of the world, we would know all we needed to ascertain the nature and existence of our concept's referent; just as we ordinarily ascertain the nature and existence of our concept's referents, contingent on how the world turns out (think of "Hesperus", "water", "Jack the Ripper"). Given this ability, we can do the same in cases of counterfactual speculation about actual-world candidates, provided that we are supplied with the relevant qualitative information. Indeed, such thought-experiments are the basic currency of the theory of reference.

We need not require that a centered world itself contains a copy of the concept. In evaluating the referent at an actual-world candidate, we retain the concept from the *real* actual world.[8] For example, in an actual-world candidate in which the individual at the center is comatose, the first intension of my concept "I", from the real actual world, picks out the automaton. Intuitively, we would say that in such a world (considered as actual), it turns out that "I am comatose" is true, just as we would say that in the XYZ-world (considered as actual), it turns out that "water is XYZ" is true. Not all concepts are so straightforward. We will see later that the first intensions of some concepts are not determinate across all centered worlds, picking out a referent only if certain background conditions are satisfied. All that matters for now, however, is that the first intension of a concept is a non-trivial partial function, determinate across a large class of worlds.

The first intension of a concept is fixed independently of any external matters in the actual world; it must be, as the intension itself specifies the way in which the referent of a concept

[7]This phrase is due to M.K. Davies and I.L Humberstone, in "Two Notions of Necessity", *Philosophical Studies* 38(1980):1-30. It is a familiar notion from two-dimensional modal logic that statements about possible worlds are evaluated differently depending on whether the worlds are considered as actual or considered as counterfactual. The basic idea is presented in G. Evans' "Reference and Contingency", *The Monist* 62(1979):161-89. For example, "water is H_2O" is true in the XYZ-world considered as counterfactual, but it is false in the XYZ-world considered as actual.

There are two corresponding modal operators, which Evans calls "deep" and "superficial" necessity. For example, "The meter stick in Paris is one meter long, if it exists" is deeply necessary but superficially contingent, as it is true in all worlds considered as actual, but false in some worlds considered as counterfactual. For "water is H_2O", the opposite is the case: this statement is superficially necessary but deeply contingent. Actual-world candidates resemble what Kaplan has called "contexts of utterance" in his theory of the semantics of demonstratives ("Demonstratives"). Counterfactual-world candidates correspond to his "circumstances of evaluation".

[8]This awkward construction is also familiar from two-dimensional modal logic (Davies and Humberstone, p. 10). This way, even when we are discussing the various worlds that might turn out to be actual, we can acknowledge that one of them in fact *has* turned out to be actual.

depends on how the world turns out. This is reflected in the *a priori* thought-experimental methodology in the investigation of first intensions. Given my concept "Gödel", I can ask (with Kripke): to what would it refer if it turned out that a man named Schmidt had proved the incompleteness of arithmetic, before the proof was stolen and published by a man named "Gödel"? Empirical knowledge about the actual world is irrelevant here, as specification of the basic facts about the centered world in question gives me all the facts that could possible be relevant. First intensions are therefore *cognitively accessible*, a point that will be important later.

The second intension of a concept works quite differently. This is determined *a posteriori*, as it depends on how the world of the thinker turns out. In evaluating the second intension at a world, we *consider the world as counterfactual*, asking: what is the reference of the concept in that world, given that the actual world has turned out as it has? For example, if we accept the intuitions of Kripke and Putnam, then given that it turns out that "water" refers to H_2O in the actual world, the second intension picks out H_2O in every counterfactual world. On the other hand, if the actual world had turned out to contain XYZ as the dominant liquid in our environment with the appropriate properties, then the second intension of my concept "water" would have picked out XYZ in all worlds instead. These counterfactual-world candidates require no center; the location of a thinker within them is irrelevant.

The full story about a concept can be summed up in a *two-dimensional intension* $F : W^* \times W \rightarrow R$, where $F(v, w)$ is the referent of the concept in w, if v turns out to be actual (for $v \in W^*$ and $w \in W$). The first intension is $f_1 : v \mapsto F(v, v')$, where $v \in W^*$ and where v' is the uncentered world obtained by "unmarking" the center of v. The second intension is $f_2 : w \mapsto F(a, w)$, where a is the actual world of the thinker. F and f_1 are determined *a priori*, and f_2 is determined *a posteriori*. Note that the first intension and the second intension always pick out the same referent in the actual world: $f_1(a) = f_2(a') = F(a, a')$.

Generally the second intension depends directly on the first intension and on the actual world, but the pattern of dependence depends on the concept involved. The simplest case is that of ordinary "descriptive" concepts— perhaps "doctor" and "circle", although semantic intuitions differ—for which the conditions of applications in a counterfactual world are the same as they are in the actual world. For these concepts, the second intension is simply a copy of the first intension, but over the space of uncentered worlds: $f_2(w) = f_1(v)$, where $v' = w$. This requires $f_1(u) = f_1(v)$ for all $u, v \in W^*$ such that $u' = v'$—that is, the value of the first intension at a centered world cannot depend on the position of the center—so the first intensions of such concepts cannot be in any way indexical. In these cases, the second intension is determined *a priori*.

More interesting for our purposes are cases in which the second intension differs significantly from the first intension, and is determined only *a posteriori*. The most straightforward way for this to happen is for concepts corresponding to rigid designators, such as indexicals and names. For such concepts, the second intension is determined by *rigidifying* the first intension—that is, by evaluating it at the actual world, and projecting that referent across all possible worlds. For

instance, it turns out that the first intension of my concept "I" picks out David Chalmers at the actual world, so the second intension picks out David Chalmers at all worlds in which he exists. A similar dependence (with minor complications[9]) holds for natural kind terms.

For a rigid designator, the structure of the second intension is quite straightforward, but characterizing the structure of a first intension is generally more complex. Often the best one can do is to give a rough summary. For instance, it is not too far off the mark to say that the first intension of my concept "water" picks out the dominant clear, drinkable liquid in the environment that stands in an appropriate causal relation to the center of a world, but this is far from perfect. One can characterize the first intension more carefully by considering difficult cases (what would the reference be if there was both XYZ and H_2O in the lakes? what would it be if the liquid that falls from the sky turns out to differ from the liquid we drink, due to a complex reaction? what if "liquids" turned out to be made of fine particles of sand? and so on), but we cannot expect to arrive at a perfect characterization of any manageable length; indeed, a perfect finite characterization may be impossible. It is sometimes useful to appeal to rough-and-ready characterizations such as "the dominant clear liquid in the environment" for the purposes of illustration, but it must be stressed that the real object of interest is always the full intension, to which any such descriptive characterization is a mere approximation.

It should be noted that this picture is entirely neutral between "causal" and "descriptive" theories of reference. On any account of how reference is fixed, there will be some way in which the referent of a concept depends on how the actual world turns out (if the actual world turns out this way, then "elm" refers to these things; if it turns out another way, then "elm" refers to those things). That is, on *any* theory of reference a concept will have a first intension. The subject matter of the theory of reference is the precise structure of first intensions; this is something about which the present account is neutral. When I occasionally characterize a first intension in descriptive terms, this is only an approximation for the purposes of illustration, and nothing will depend on the details.

The framework applies equally to concepts that are used with *semantic deference*. As Putnam and Burge have pointed out, my concepts of "elm" and "arthritis" can refer appropriately despite my knowing very little about the relevant subject matter. In these cases, I refer via the division of linguistic labor. At a first approximation, if those around me (and especially the "experts" in my community) refer to X-trees by "elm", then my concept "elm" refers to X-trees; if they refer to Y-trees by "elm", then I refer to Y-trees, and so on. The concept therefore determines a perfectly good first intension. To characterize this first intension even approximately, we have to involve the *name* of the concept: thus, at a first approximation, my concept "elm" picks out whatever trees those around me (particularly experts) refer to as "elms". As always, reference

[9]Strictly speaking, natural-kind concepts are not rigid designators: "cat" does not pick out the same referent at every world, as there are different cats in each. One way around this is to stipulate that "cat" picks out the property of cathood at every world. Less artificially, we can give a two-stage story according to which "cathood" picks out the same property in each world, and "cat" picks out the objects satisfying the property in that world. I will ignore these complications in what follows.

of a concept depends on how the actual world turns out; it is just that in these cases, some of the relevant facts about the actual world are linguistic.

Putnam suggests that concepts like "water" and "elm" show that the intension of a concept cannot determine the extension, if an intension is internally determined. The two-dimensional analysis shows that this is only half the truth. The first intension of a concept determines its extension, and the first intension is internally determined. Of course, the first intension is a *centered intension*; it takes a centered world as argument, so it may not strictly falsify Putnam's claim. But any intension requires facts about the actual world to determine extension; and it is most natural to regard the actual world of a thinker as centered. So it is no surprise that centered intensions should be central in the fixation of reference.

In some ways, the first intension of a concept resembles a Fregean sense. It is the first intension that fixes reference in the actual world, and the first intension is generally cognitively accessible. Instances of Frege's puzzle—the informativeness of identity statements such as "Hesperus is Phosphorus"—arise precisely when the concepts involved have different first intensions. Insofar as senses are introduced to solve these problems, first intensions can play the role. On the other hand, Frege required that senses determine reference non-indexically, stipulating that the sense of a concept such as "here" is tied to a particular place. Here, the second intension seems to be a closer fit. Many have noted that nothing can satisfy all Frege's requirements on sense,[10] and it is perhaps fruitful to view the present account as dividing the notion of sense into two distinct components. A case can also be made, though, that the first intension is a sufficiently close fit that it or something like it should be identified with a concept's sense. It is not implausible to maintain that the requirement of non-indexicality is misguided, and that a first intension can do much of the important work that senses are introduced to do, despite this and other differences of detail with Frege's account.

There should be no question of whether the first or second intension is *the* intension associated with a given concept. The full story can only be given two-dimensionally. One or the other may be more useful for various specific purposes. In matters of linguistic content, the second intension may play the major role, as different users of a term may have quite different associated first intensions, whereas the second intension will generally be constant. This is particularly clear in the case of names, for which first intensions can vary wildly across a community. For questions about thought and its role in governing action, however, we will see that the first intension plays a more central role.

3 Narrow and Wide Content

So far we have been concerned with the semantics of *concepts* (the mental analogs of terms). Now we move to the central topic, the semantics of *thoughts* (the mental analogs of sentences).

[10] See especially J. Perry, "Frege on Demonstratives", *Philosophical Review* 86 (1977): 474-97.

As before, I will be concerned with thoughts (usually beliefs) that are associated with sentences in the thinker's language, but the account generalizes.

Just as concepts are associated with intensions, thoughts are associated with *propositions*, which here will be regarded as functions from possible worlds into truth-values, or more simply as sets of possible worlds. (More "fine-grained" accounts of propositions can be developed compatibly with the current intensional framework—for example, propositions can be seen as a structured group of intensions—but the coarse-grained account is all we will need here. The framework below can be straightforwardly applied to a more fine-grained account.) Propositions be regarded as the contents of thoughts and of sentences, just as intensions can be regarded as the contents of concepts and of terms.

Just as there are two intensions associated with every concept, there are two propositions associated with every thought. The *first proposition* is a *centered proposition*—a set of centered worlds—and is precisely that set of centered worlds such that the belief would be true if that world turned out to be actual.[11] For instance, the first proposition associated with my thought "I am a philosopher" is the set of centered worlds in which the individual at the center is a philosopher. The second proposition, on the other hand, is the set of counterfactual worlds of which the belief, as thought in the actual world, is true. The second proposition associated with the thought above is the set of worlds in which David Chalmers is a philosopher.

The first proposition associated with a thought can usually be determined by composing the first intensions of the concepts involved.[12] For example, the first proposition of a thought "Hesperus is Phosphorus" will be true at a centered world if the individual picked out by the first intension of "Hesperus" in that world is also picked out by the first intension of "Phosphorus" in that world—that is, roughly at those worlds in which the morning star is the evening star. A similar relation holds between second intensions and second propositions. The second intensions of "Hesperus" and "Phosphorus" pick out the same object in all worlds in which the referent of either exists, so the second proposition of "Hesperus is Phosphorus" is true of all such worlds.

To determine the first proposition associated with my thought, I consider a particular centered world, and ask: would my thought be true if that world turned out to be actual? Consider my thought "There is water on the table in front of me". If the actual world turned out to contain

[11] In "Assertion", Stalnaker introduces a proposition like this as the "diagonal proposition" associated with an utterance. Just as the first intension can be seen as the "diagonal" of a two-dimensional intension— $f_1(v) = F(v, v')$—the first proposition can be seen as the diagonal of a "two-dimensional proposition", or a function from $W^* \times W$ into truth-values. Perry (in "Cognitive Significance and the New Theories of Reference", *Nous* 22(1988): 1-18) introduces a closely related proposition as the proposition "created" by an utterance, as opposed to the proposition "expressed" by an utterance (which is the second proposition).

[12] Note that nothing here requires a "language of thought". These considerations about the semantics of thoughts are independent of questions about how thoughts are physically realized: thoughts might have a compositional semantics even if they lack a compositional syntax. I will usually assume a compositional semantics for thoughts in this discussion, but it should be noted that the central part of this account does not strictly require even this. The definition of first and second propositions requires only that thoughts are the kind of thing can be true or false of the world.

XYZ as the dominant liquid in the oceans and lakes, and if there were a dining table in front of me supporting a glass of XYZ, then my belief would be true. As things really are, the world contains H_2O in the oceans and lakes, and there is only an empty glass on the study table in front of me, so my belief is false. At a rough approximation, the first proposition associated with my thought is the set of centered worlds in which there is a sample of the dominant clear liquid in the environment on a table in front of the individual at the center. (As usual, the rough approximation is no substitute for the proposition itself, which can be obtained only from careful consideration of detailed cases.)

The second proposition associated with my thought is entirely distinct. Given that the actual world turns out as it has, with H_2O in the oceans, my study table T in front of me, and with my thought occurring at time t, the second proposition is that set of worlds in which there is a sample of H_2O on table T at time t. As is often the case, the second proposition can be characterized only *a posteriori*, as it depends on the environment of the thinker.

I will call the first proposition associated with a thought the thought's *notional content*, because it most directly reflects how things seem from the point of view of the thinker. I will call the second proposition the thought's *relational content*, as it generally depends on the relations that the thinker bears to objects in the external world. For example, when you think "I am hungry" and I think "You are hungry", our thoughts have different notional content, but similar relational content. When we both think "I am hungry", the situation is reversed.

Importantly, the notional content of a thought is determined entirely by the internal state of the thinker.[13] Like the first intension, the first proposition cannot depend constitutively on the external world, as the proposition itself specifies how the truth of a thought depends on how the actual world turns out. I can investigate the first proposition associated with my thought without investigating the environment; all I need to do is ask, *if* the environment turns out like so, will my thought be true? This is an entirely internal enterprise; facts about the actual environment have no role to play. (Of course, the notional content of a thought will almost always depend *causally* on the external world. The external world plays a crucial role in the genesis and dynamics of the notional contents of our thoughts; but it always does so by affecting the internal state of the thinker.).

Every thought has both notional and relational content. As notional content is internally determined, we may stipulate that a thought's *narrow content* is its notional content.[14] Similarly, we can stipulate that a thought's *wide content* is its relational content. It should be noted, however, that relational content is not always dependent on the environment. In particular, if a thought involves only non-rigid concepts used without semantic deference, the associated

[13]Thus even a brain in a vat might have thoughts with notional content. This can be used to defuse Putnam's anti-skeptical argument that if he were a brain in a vat, he could not think "I am a brain in a vat" (*Reason, Truth, and History*, Cambridge: Cambridge University Press, 1981). A brain in a vat could think a thought with the appropriate notional content, if not the appropriate relational content, and the notional content of the thought would seem sufficient to express a significant skeptical possibility, true only in worlds in which the individual at the center lacks the usual sort of epistemic contact with the surrounding world.

second proposition will be a simple twist on the first proposition, so that relational content will be internally determined.

As promised, this sort of narrow content is truth-conditional. The notional content of a thought delivers the conditions a centered world must satisfy, in order for the thought to be true in that world. The notional content of my thought "There is water on the moon" is true in a centered world with XYZ in the oceans and lakes and XYZ on the moon, even though the relational content of my thought is false there. At a rough approximation, the notional content will be true in those centered worlds in which the dominant clear liquid found in the oceans and lakes surrounding the center can also be found on the heavenly body bearing the appropriate relation to the center.[15] We can think of the notional truth-conditions of a thought as expressing the way the actual world has to be, in order for the thought to be true.

The notional truth-conditions of a thought are often very different from the relational truth-conditions. One might worry that a thought might therefore turn out to be both true and false, but this is impossible. Just as the first and second intensions pick out the same referent in the actual world, the first and second propositions share the same truth-value there, so there is no danger of a thought's being both true and false in actuality. It is only at other worlds that the truth-conditions will give different results, and this is a consequence of the way those worlds are treated. In evaluating the first proposition they are considered as actual, whereas in evaluating the second proposition they are considered as counterfactual. We might say that notional truth-conditions tell us what it is for a thought to be true *in* a world, whereas relational truth-conditions tell us what it is for a thought to be true *of* a world.

4 The Advantages of Notional Content.

In recent times, the "content" of a thought has generally been identified with its relational content; but the notional content seems to be an equally good candidate. As before, there is no need to decide which is *the* content. Once we have acknowledged the two-dimensionality of meaning, the two kinds of content can comfortably co-exist. The "content" of a thought decomposes naturally into its relational and notional content. That being said, there are many

[14]This account of narrow content is closely related to that suggested by Lewis, who identifies the content of a belief with a self-ascribed property, or an "egocentric proposition" (see "Attitudes *De Dicto* and *De Se*" and "Reduction of Mind").

[15]The distinction between notional and relational truth-conditions parallels Kripke's distinction between the *a priori* and the necessary. A thought is knowable *a priori* if its notional content is true in all centered worlds; and it is necessarily true, in Kripke's sense, if its relational content is true of all worlds.

A related two-dimensional account can be given of the corresponding phenomenon in language, yielding in a certain sense a semantic explanation of *a posteriori* necessity. I develop such an account in my *Toward a Theory of Consciousness* (forthcoming from MIT Press), where it is used to analyze some applications of *a posteriori* necessity in the mind–body problem. Upon analysis, the two-dimensional account suggests that some of the more ambitious applications of *a posteriori* necessity (in defence of certain sorts of materialism and of certain sorts of moral realism, for instance) are unlikely to succeed.

ways in which the notional content of a thought is responsible for most of the work that we would expect a notion of "content" to do.

First, the notional content of a conscious thought better reflects the way the thought presents itself to its thinker. The relational content of my thought that there is mercury in the cabinet, for instance, will be true of only those worlds in which the cabinet contains an element with a certain atomic number, but I have no idea what that number is. The second intension of "mercury" and the relational content of the thought are neither directly apparent nor useful to me. The first intension of "mercury", by contrast, picks out something like the silvery liquid called "mercury" that is found in thermometers; this much more closely parallels the phenomenology of the concept. The notional content of my thought holds in roughly those worlds in which the cabinet contains such a liquid. This more directly reflects what it seems to me that I am thinking, when I am thinking that there is mercury in the cabinet. Of course, even notional content is not always transparent to the thinker, as possibilities such as unconscious thought and self-deception demonstrate, but it is more accessible than relational content.

Secondly, notional content determines the *cognitive* relations between thoughts. If one thought implies another thought *a priori*, it is because the first proposition associated with the first implies the first proposition associated with the second. If I know that it is hot where I am now, I know that it is hot here, and vice versa; this is reflected in the fact that the notional contents of the two thoughts are the same. The relational contents of the thoughts are very different, however: there is no obvious relation between the proposition that it is hot where David Chalmers is at time t and the proposition that it is hot at place P then.

It is straightforward to see why this is so: if one thought implies another *a priori*, then in any centered world in which the first turns out to be true, the second will turn out to be true, so the notional content of the first will imply the notional content of the second. Conversely, if the notional content of one thought implies the notional content of another, a thinker ought to be able to infer one from the other *a priori* (setting aside problems with complex mathematical statements and the like), as the relationship between the thoughts is independent of facts about the external world. This is not so for relational content: the relational content of a thought may depend on facts about the external world that are unknown to the thinker, so that relations between thoughts in virtue of their relational content may lie outside the realm of cognitive access. As with first and second intensions, notional content is cognitively accessible in a way in which relational content is not.

This can be applied straightforwardly to explain the informativeness of a statement such as "Hesperus is Phosphorus". Although its associated second proposition is equivalent to that of the trivial "Hesperus is Hesperus", its associated first proposition is quite distinct, so it is not cognitively trivial. If a thought is notionally equivalent to a thought that is knowable *a priori*, it will itself be knowable *a priori*, but an equivalence in relational content carries no such guarantee.

We can also invoke notional content in the case of Kripke's Pierre, who in France heard that "Londres est jolie" and believed it, then arrived in London and thought it was ugly, never

suspecting that "London" and "Londres" named the same city.[16] Paradoxically, Pierre seems to simultaneously believe that London is pretty and that London is not pretty, without any breakdown in rationality. But the two-dimensional framework shows why there is no paradox. Pierre's concepts "Londres" and "London" have quite different first intensions—in a given centered world, the first picks out (approximately) the famous city called "Londres" that the individual at the center has heard about, whereas the second picks out the grimy city in which that individual has been living. The second intensions are identical, of course, picking out London in every world. Pierre's two beliefs, then, have contradictory associated second propositions, but their first propositions are compatible. Contradictory second propositions support no charge of irrationality, as rational relations are determined by the first dimension of content.

Intuitively, Pierre's two beliefs are rationally compatible because there is some way the world could turn out such that both are true. There is a centered world in which "Londres" names a faraway, beautiful city (maybe it is in India), and in which the individual at the center inhabits an entirely distinct ugly city called "London"; and for all Pierre knows and believes, such a world could be actual. As long as there is such a world, satisfying the notional content of all Pierre's thoughts—that is, as long as the *notional* contents of his thoughts are compatible—his rationality is not in danger.

This brings out the relation between this account and a suggestion of Dennett's[17] —that the narrow content of a thought is reflected in the *notional world* of the thinker. The notional world we can take to be a world (really a class of worlds) in which all of a thinker's beliefs (or as many as possible, in cases of irrationality) would be true. A notional world is to be thought of as a world as the thinker finds it, so of course it is a centered world; and beliefs are to be evaluated in that world according to their first propositions. Pierre's notional world is a world in which there is a beautiful faraway city called "Londres", and a grimy city close at hand called "London". If Pierre really lived in his notional world, he would be right about everything and rarely surprised.

A third advantage of notional content, and perhaps the most important, is its suitability for a role in the explanation of behavior. It is often noted that relational content seems slightly out of synchrony with what one would expect of an explanatory psychological state. To use an example of Perry's,[18] if you are watching me and my pants catch fire, our respective beliefs that my pants are on fire now will have the same relational content (they are true in all worlds in which David Chalmers' pants are on fire at time t), but will lead to very different actions (I might jump into a river, while you just sit there). The difference between our actions does not seem to be something that a characterization in terms of relational content alone can explain. In a similar way, belief states can produce very similar behavior for apparently systematic reasons, even when the beliefs have very different relational content—witness the behavior that my twin and I produce when we think about twin water and water respectively, or the similarity between

[16] S.A. Kripke, "A Puzzle about Belief", in *Meaning and Use*, ed. A. Margalit (Dordrecht: Reidel, 1979).

[17] D.C. Dennett, "Beyond Belief", in *Thought and Object*, ed. A. Woodfield (Oxford: Oxford University Press, 1981).

the actions of two people who think "I am hungry". A whole dimension of the explanation of behavior seems out of the reach of relational content.

These explanations can be easily handled in terms of notional content. If you and I think "I am hungry", the notional contents of our thoughts are very similar—our thoughts are associated with similar first propositions, although different second propositions—and that similarity is reflected in the similarity of our actions. When you and I both believe that my pants are on fire, on the other hand, our notional contents are very different, and our actions differ correspondingly. The Twin Earth case can be handled similarly. Even though the thoughts of my twin and I refer to XYZ and H_2O respectively, their associated first propositions are the same, and this similarity in notional content is reflected in the similarity of our actions.

Narrow content is often postulated to account for the similarity in behavior between twin cases, but we need not move to the realm of science fiction to see the point. Two thoughts can share narrow content even when two thinkers are quite different, as our thoughts "I am hungry" show, and even in these cases, similarities in narrow content will lead to similarities in action, other things being equal. Say I think that Superman is across the road, and I want to have Superman's autograph; then other things being equal, I will cross the road.[19] If you have thoughts with similar notional content to mine, then you will do the same. But if your thoughts only share relational content with mine, but have different notional content—say you think that Clark Kent is across the road, but want Superman's autograph—then your behavior may be correspondingly quite different.

In general, whenever the content of a thought is apparently relevant to behavior, its contribution is "screened off" by that of notional content in the following sense: if the thought had had the same notional content but different relational content, the behavior would have been the same, whereas if it had had the same relational content but different notional content, the behavior might have been quite different.

To see the latter point, we need only examine cases like those above. The thoughts "I am hungry" and "The guy over there is hungry" (unknowingly looking in a mirror) will lead to very different behavior, even though their relational content is the same. When Lois Lane is trying to cut Clark Kent's hair, her observation "Clark's hair breaks the scissors" will prompt a reaction very different from that provoked by a corresponding thought concerning Superman. If I hear that Cary Grant is starring in a movie, I might be more likely to watch than if I hear that the movie stars Archie Leach. In all these cases, different reactions are provoked by a difference in the notional content of a thought. In general, whenever the notional content of a thought is varied,

[18]These examples are discussed in Perry's "The Problem of the Essential Indexical", *Nous* 13 (1979):3-21. Perry observes that beliefs, insofar as they play a role in the explanation of behavior, are "essentially indexical", and wonders how this can be squared with a propositional view of belief content. This indexicality is explained on the current account by the facts that (1) notional content, not relational content, governs behavior, and (2) notional content, unlike relational content, is a centered proposition.

[19]To simplify the discussion, I make the happy assumption that Superman is actual and is identical to Clark Kent.

different consequences can be expected, even if relational content is preserved throughout.[20] After all, we have already seen that notional content governs cognition, and cognition governs action.

By contrast, if the relational content of a thought is varied but notional content is kept constant, behavior is preserved throughout. Perhaps, unbeknownst to me, Cary Grant is an elaborate hoax, a co-operative construction by avant-garde animators and the mass media. In such a case, my thought about Cary Grant will have no relational content at all, but as long as it has the same notional content my behavior will be the same. Or perhaps Cary Grant is really Ludwig Wittgenstein in disguise: if so, very different relational content, but the same behavior. Similarly, when my twin and I think "I need some more water for this pot", the relational contents of our thoughts differ, but we both rush for the sink. As usual, notional content does all the work.

It should be noted that in making this case for the primacy of notional content, we have not needed to appeal to any *a priori* methodological principles such as the dictum that what governs behavior is in the head. The primacy of notional content can be seen on grounds entirely independent of questions about how content is physically realized. Still, such considerations can further illustrate the case. When two thinkers have thoughts with the same notional content but different relational content, this will usually be due to a difference in features of the thinkers' environment. Usually the relevant features will not even be features of the thinkers' proximal environment, but rather features of the speakers distal, historical environment (this applies to almost all non-demonstrative concepts: think of "water", "arthritis", "Cary Grant"). Such features seem much too far removed to play any direct role in the causation of action.

The primacy of notional content runs deeper than the internal/external divide, however. Even in cases where differences in relational content are due to factors "in the head", relational content is still irrelevant to action. Taking an example of Evans',[21] let "Julius" be a descriptive name for the inventor of the zip: its first intension picks out whoever invented the zip in the actual world, and its second intension picks out that person in all possible worlds regardless of whether they invented the zip there. Then the second intensions of "Julius" and "The inventor of the zip" are very different. But it is clear that any thoughts "Julius is P" will play the same role in directing cognition and behavior as thoughts "The inventor of the zip is P". This is precisely because of the similarity in the notional content of the thoughts. The differences in second intensions, and the corresponding differences in relational content, are entirely irrelevant.

Why is notional content primary? To answer this question, it is useful to think of my belief contents as constituting a model of my world, a kind of map by which I steer. My model is

[20] Of course, thoughts like "Cary Grant is in the movie" and "Archie Leach is in the movie" might lead to the same actions despite their different notional content, if I know that Cary Grant is Archie Leach. But even here, there are some circumstances under which the thoughts might play a different role—if someone tells me that Cary Grant is not Archie Leach after all, for instance. In general, whenever two thoughts have different notional content, there are at least hypothetical circumstances under which the action-governing roles of the thoughts will differ.

[21] "Reference and Contingency".

a model of the world as I find it, a centered world with me at the center, and my beliefs are constraints on that world. One belief might rule out *these* centered worlds as a candidate for the world where I am, another might rule out *these*, until only a limited space of worlds is left. I operate under the assumption that my world is one of those worlds, and if I am lucky I will not be too surprised.

My world-model is ultimately a *notional* world: a set of centered worlds, such that none of these would overly surprise me if they turned out to be actual. The constraints on this model are those of notional content. Relational content puts some additional constraints on the world, but these are not constraints that are useful to me. The relational content of my belief that the liquid in thermometers is mercury endorses only those worlds in which thermometers contain the element with atomic number X, but this constraint is so deeply inaccessible to me that if it turned out that the liquid has atomic number Y, I would not be in the least surprised. In an important sense, this constraint is not reflected in my world-model at all. Insofar as my world-model is useful to me, the constraints on it are entirely those of notional content; and insofar as content is useful to me, it is notional content that carries the burden.

It should be stressed that nothing here implies that facts about a thinker's environment are irrelevant to the explanation of behavior. Such a claim would be implausible. For example, facts about environmental history will be invoked in a causal explanation of why a thinker comes to have certain beliefs; facts about the current environment are crucial to explaining the success or failure of various actions; and facts about the proximal environment clearly play an important role insofar as they affect the thinker. All that follows from the present framework is that *relational content* is redundant in the explanation of behavior. Facts about the environment are relevant to explanation in all sorts of ways, but they are not relevant in virtue of their role in constituting content. The kind of *content* that governs behavior is purely internal.

5 Belief Ascription and Psychological Explanation

This raises something of a puzzle about the role of belief ascriptions in psychological explanation. If what has gone before is correct, the kind of content that governs cognition and action is notional, and therefore narrow. But at the same time, there is strong evidence that the kind of content attributed by belief ascriptions is at least sometimes wide. Does this mean that the common-sense framework of explanation in terms of belief ascription should be discarded? Alternatively, is the apparent success of the common-sense framework evidence that something in these arguments has gone terribly wrong?

Neither conclusion is justified. The present framework shows how it can at once be true that (1) belief ascriptions ascribe wide content, (2) narrow content governs action, and (3) belief ascriptions explain action. In short: Belief ascriptions ascribe a combination of notional and relational content. It is in virtue of the relational component that the ascribed content is wide, and it is in virtue of the notional component that the ascribed content is explanatory.

A full justification of this answer requires two things: first, an analysis of what is attributed in belief ascriptions, so that we can see precisely what sorts of notional and relational content are attributed, and second, an analysis of the role of belief ascriptions in psychological explanation, so that we can see that even in ordinary practice, it is the notional aspect of content that carries the explanatory burden. I cannot provide anything like a complete treatment of these matters—the analysis of belief ascriptions deserves entire volumes of its own—but I can provide a preliminary sketch.

It is easy to see that ordinary belief ascriptions ascribe both notional and relational content. If I say "Ralph believes that Clark Kent is muscular", in order for my utterance to be true Ralph must have a belief that satisfies two sorts of constraints. First, the belief must have the relational content of the proposition that Clark Kent is muscular (perhaps we can allow a certain amount of variation in the relational content, if for example his concept of muscularity is slightly different from the norm). But that alone is not enough: a belief that Superman is muscular would have the same relational content, but would not make my ascription true. For the ascription to be true, his belief must involve a concept that refers to Clark Kent under an appropriate first intension: perhaps an intension that picks out whoever is called "Clark Kent", or one that picks out whoever is that reporter with glasses at the Daily Planet, or more likely some complex intension in the vicinity.[22] A variety of first intensions will do, in order for the ascription to be true, but still a narrow class compared to all the first intensions that in fact refer to Clark Kent. A rigidified concept whose first intension picks out the guy in the cape, or the strongest man in the world, might refer to Clark Kent, but the thought that the person in question is muscular would not make my ascription true.

More straightforwardly, if I am right in saying "Tom believes that he is hungry", then Tom must have a belief with more or less the appropriate relational content, true of all those worlds in which Tom is hungry at time t, but there is a strong constraint on notional content too. In particular, Tom must refer to himself via the first intension that picks out the individual at the center in every centered world. If he sees someone in the distance clutching their belly, without realizing that he is in fact looking into mirrors, then a thought that that person is hungry has the right relational content, but on the most natural reading it does not make my ascription true. The ascription will only be true if Tom's belief refers to himself *as* himself; that is, only if its notional content is appropriate.

We have seen that content decomposes naturally into notional and relational content; we now see that belief ascription puts strong constraints on both. The precise nature of these constraints is the subject matter of a theory of belief ascription. Ideally, such a theory will yield, for any given ascription, the conditions that a beliefs' notional and relational contents must satisfy in

[22]Thus the first intension of a concept is a natural candidate to play the role of a "mode of presentation" in contemporary theories of the semantics of belief ascriptions. The possibility seems to have been widely overlooked: for example, it is not mentioned in S. Schiffer's otherwise thorough survey of potential modes of presentation in "The Mode-of-Presentation Problem", in *Propositional Attitudes*, ed. C.A. Anderson and J. Owens (Stanford: Center for the Study of Language and Information, 1990).

order to make the ascription true. We can think of a belief ascription as marking out a subspace in the space of (notional content, relational content) ordered pairs.

The constraints on relational content are generally fairly straightforward, as the examples above show. Usually, these constraints can be derived from the reference or meaning of the terms involved in the ascription. The constraints on notional content are vaguer and more subtle. Certainly not every first intension referring to Clark Kent qualifies a concept to play a role in making a "Clark Kent" belief ascription true, but the precise restrictions on what counts as a "Clark Kent" first intension are not at all obvious. Similarly, to believe that water contains hydrogen atoms, it presumably does not *suffice* to have a relevant belief about H_2O (imagine a very theoretical chemist who *only* knows about H_2O's chemical properties); a "water"-appropriate first intension is required, but just what are the conditions here? This is a matter for careful linguistic investigation. The constraints on notional content will be somewhat indeterminate in intermediate cases, and they will probably be context-sensitive,[23] but they will be there, and they will generally be quite strong.

Given that notional content governs action, it follows that if belief ascriptions are to causally explain action, it must be in virtue of the notional content ascribed; the relational content ascribed is redundant to the explanation. To make this case more directly requires an examination of many specific examples wherein belief ascriptions are used to explain action, but the general point can be straightforwardly illustrated. One way to see the primacy of notional content is to consider beliefs involving empty names, such as "Santa Claus". These beliefs have no non-trivial relational content (statements about Santa Claus are either false or lack truth-value in all worlds), and no non-trivial relational content is ascribed by the corresponding ascriptions, but ascription of beliefs about Santa Claus seem to function in *precisely* the same way in the explanation of action as do ascriptions of beliefs about real people. We might explain young Karen's agitation on Christmas Eve in terms of her belief that Santa Claus is coming, that he will not fit down the chimney, and so on. Santa's non-existence and the corresponding absence of relational content make little difference to the success of such an explanation. What governs Karen's actions are her *notions* of Santa Claus; and what governs the success of the explanation is the notional content that these ascriptions ascribe to her. And this is a *typical* case of the role of belief ascriptions in explanation: even when non-trivial relational content is ascribed (as when the referent of the name exists), it makes little difference to the patterns of explanation.

In every case in which content explains action, we can see that the explanation succeeds even if the relational content attributed is ignored. For instance, if we explain my opening the refrigerator in terms of my belief that there is water in the refrigerator and my desire for a glass of water, we never need to invoke the H_2O-involving relational content. The explanation gains sufficient purchase from the notional content ascribed alone—roughly, the content that there is some of the clear liquid with the appropriate properties in the refrigerator, and that I want some

[23] As suggested by M. Crimmins and J. Perry, "The Prince and the Phone Booth: Reporting Puzzling Beliefs", *Journal of Philosophy* 86 (1989): 685-711.

of that liquid, and so on.

It might be objected that there are cases in which the constraints on the notional content ascribed by a belief ascription are weak, so that relational content must be doing any explanatory work. I think that ascriptions putting weak constraints on notional content are rare, but assume they occur—perhaps an attribution of a belief about Smith constrains the relevant first intension very little. Even so, if we look at how such an ascription can be used in explanatory practice, we see that notional content is still doing the real work. For example, perhaps we explain why Bev goes to the pub by saying that she wants to see Smith and believes that Smith is at the pub. Leaving aside constraints in the concepts of seeing, the pub, and so on, there is even a constraint on notional content implicit in the "Smith" attributions. For notice: it is implicit here that the two "Smith" concepts in Bev's thoughts have the same first intensions. If her belief and her desire had very different notional content where "Smith" is concerned—perhaps she wants to see Batman and believes that Smith is at the pub, not knowing that Smith is Batman—the inference from those states to her action would fail. So there is a strong joint constraint on notional content: despite a lack of constraint on the individual beliefs, Bev is implicitly ascribed the belief that a person she wants to see is at the pub. It is *this* ascribed belief that is doing the real explanatory work, and this ascription clearly puts a heavy constraint on notional content. To make the case that all such examples can be similarly analyzed requires a detailed treatment, but this illustrates the general pattern.

One might wonder why relational content is ascribed at all, given its explanatory idleness. I think there are two answers, both tied to language. First, given that we ascribe beliefs in the same language we use to describe the world, it is natural to do so in a world-involving way. Our languages make relational content much easier to directly ascribe than notional content. Second, relational content is important to understanding the success of communication and of collective action. When I tell you that I have a cold, you acquire a thought whose notional content is different from mine, but whose relational content is the same. Communication very frequently involves transmission of relational content, and our collective cohesion (if not our individual actions) can often be understood in terms of shared relational content. But both of these points deserve a much more extensive development.

In any case, we have seen that the centrality of narrow content in the causation of action need not overthrow the role of belief ascriptions, as some have suggested it should. At most we have shown that belief ascriptions are a somewhat rough-edged tool, due to the way they wrap both components of content into a single parcel, bringing the idle relational content into play alongside the notional content that does all the work. But this should not surprise us; we cannot expect a folk theory to be maximally efficient.

In moving from common-sense psychology toward a developed cognitive psychology, we might expect that the kind of content that is invoked will become more purely notional, and that relational content will be relegated to a secondary role or dropped entirely.[24] We might also expect that better tools will be developed to specify the notional contents of thoughts than the current rough-and-ready language of belief ascription. This might qualify as a revision of our

folk notion of belief, emphasizing and refining the notional elements that are already present within it. But precisely because those elements are already present and playing a central role in our practice, such a development would fall well short of elimination.

6 Related Suggestions and Some Objections

The two-dimensional framework developed here bears an obvious similarity to the framework developed by Kaplan for the semantics of demonstratives.[25] Kaplan takes a term's second intension—a function from circumstances of evaluation into extension—as its "content", and defines its "character" as a function from context of utterance into content. He gives a similar treatment for the content and character of a statement. Kaplan's circumstances and contexts play the same role as our uncentered and centered worlds, and something like a first intension can straightforwardly be recovered from character, by evaluating character in a centered world and evaluating the resulting content in the same world. Kaplan's theory is an account of linguistic meaning rather than thought content, however, and at least on this account a name or a natural-kind term has the same content in *any* context of utterance (the term "water" uttered on Twin Earth is simply a different word with a different meaning), so it does not generalize to handle problems about the narrow content of thought.

The appropriate generalization has been made by Fodor[26] (and in a related suggestion by White[27]), who defines the content of a thought the usual way (as relational content), and defines the narrow content—or partial character —of a thought as a function from contexts to content. (Unlike the term "water", a "water" thought can have different contents in different contexts.) This yields a formalism very similar to the present one, but with the crucial difference that by making relational content primary, narrow content becomes entirely derivative and its independent *semantic* function is lost. Fodor says as much:

> Indeed, if you mean by content what can be semantically evaluated, then what my water-thoughts share with Twin "water"-thoughts isn't content. Narrow content is radically inexpressible, because it's only content *potentially*; it's what gets to be content when—and only when—it gets to be anchored. We can't—to put it in a nutshell—*say* what Twin thoughts have in common. This is because what can be

[24] It is not implausible to argue that cognitive psychology is already most concerned with notional content rather than relational content, insofar as it is concerned with content at all. For example, the psychological literature on concepts seems to be largely concerned with how concepts are applied to the actual world, concentrating on something like the first intensions of the concepts involved. See E.E. Smith and D.L. Medin, *Categories and Concepts* (Cambridge, MA: Harvard University Press, 1981), and S. Patterson, "Individualism and semantic development", *Philosophy of Science* 58:15-35.

[25] "Demonstratives".

[26] *Psychosemantics*.

[27] "Partial Character and the Language of Thought".

> said is ipso facto semantically evaluable; and what Twin-thoughts have in common is ipso facto not.[28]

Fodor and White nowhere suggest that narrow content has truth-conditions of its own. Despite the formal similarities, they appear to have missed the crucial point that by making narrow content a function from context into *truth-value* (or from context into referent, for concepts), narrow content becomes semantically evaluable in its own right, along an independent semantic dimension. It is a very small difference, but one that is entirely responsible for the explanatory power of the current framework.

Another suggestion that is closely related to the present one has been canvassed occasionally.[29] This is the suggestion that the narrow content of a term like "water" corresponds to a surface-level description of it, or a list of its apparent properties, such as "the clear, potable, thirst-quenching liquid found in the oceans and lakes". But this strategy suffers from a fatal flaw, pointed out by many:[30] The resulting content is not narrow but wide. The content of a term like "liquid", for instance, may be bound to the empirical internal structure of liquids. We can construct a twin case, where liquids are replaced by superficially identical masses of fine grains of sand. Intuitively, my twin does not share even the belief content above with me, as his beliefs are not about *liquids* at all.

It might be thought that this criticism applies to the present approach. After all, I have said that the first intension of "water" picks out approximately the clear liquid with the appropriate properties. But importantly, this is only to get a rough handle on the intension for the purposes of illustration. The real narrow content is an abstract function from centered worlds into extensions, and can only imperfectly be captured by a given description. We can only characterize the intension fully by detailed consideration of its value at specific worlds. As soon as we move to a summarizing description in language, imperfections are introduced, and the narrowness of the content is impurified. But the intension itself remains narrow; we should not mistake the linguistic description for the real thing.

Something very like the present proposal for the narrow content of a thought has been canvassed in a paper by Stalnaker that is generally *critical* of the idea of narrow content.[31] Stalnaker takes wide (relational) content to be primary, but defines the "diagonal" proposition associated with a thought as the function that evaluates a thought's wide content in its own context, yielding a truth-value, and suggests that the diagonal proposition might be what advocates of narrow content have in mind. The diagonal proposition is of course almost precisely the first proposition, which I have suggested is the notional content of a thought.

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[28] *Psychosemantics*, p. 50.

[29] The suggestion can be found in Fodor's unpublished "Narrow Content and Meaning Holism".

[30] E. LePore and B. Loewer, "Solipsistic Semantics", in *Midwest Studies in Philosophy* 10 (1986):595-614; K. Taylor, "Narrow Content Functionalism and the Mind-Body Problem", *Nous* 23 (1989):355-72; White, "Partial Character and the Language of Thought".

[31] R. Stalnaker, "Narrow Content", in *Propositional Attitudes*, ed. C.A. Anderson and J. Owens (Stanford: Center for the Study of Language and Information, 1990).

In his brief but rich discussion, Stalnaker makes three objections to this construal of narrow content, all of which are worth addressing. The first is as follows.

> First, this explanation makes clear that the determination of narrow content presupposes that we can identify a thought independently of its content. [...] It is not at all clear in the case of beliefs, intentions, and other states and attitudes that one can identify something that is the belief and intention in abstraction from its content, something about which we can ask, what would the content of *that* belief have been if *it* had been a belief I had on twin earth.[32]

This is surely misguided. If we could not fix reference to a thought by something other than its content, it would be impossible for us to have non-trivial knowledge of any thought's content; any knowledge that a thought has a given content would be trivial, like our knowledge that Julius is the inventor of the zip. In practice, we identify mental states in all sorts of ways—as something like *this* belief, or the belief expressed by such-and-such an utterance—and then we notice how they behave under evaluation. We can evaluate a thought in worlds one way, and thus determine its relational content; and we can evaluate the thought in centered worlds another way, thus determining its notional content. The situation with respect to the two kinds of content are quite symmetrical. It is even possible for us to fix reference to a particular belief by wide content, and then investigate its notional content. For example, we can talk about "Pierre's belief that London is pretty", and ask how its truth-value and relational content might vary in different centered worlds.

Someone might stipulate that relational content is *essential* to a thought, so that we cannot talk of how a thought's relational content might vary in different centered worlds; but this trivializes the case against notional content. It is clear that there is a sense of the word "thought" in which it is reasonable to say "If the actual world turned out to be the XYZ-world, my "water"-thought would be about XYZ", and this is the sense we are using. There is clearly something in the vicinity of a thought that can be evaluated differently in different centered worlds; whether this or something else gets to be called a "thought" is not a real issue.

Stalnaker's second objection is the following:

> Even if these questions receive satisfactory answers, it will remain true that the narrow content or realization conditions of a thought, explained as I have explained them, will be defined only relative to a very limited set of possible worlds: specifically, only for possible worlds containing the thought token. The realization conditions for the thought are satisfied in a given possible world if and only if the proposition expressed by the thought *in that possible world* is true. Suppose the thought I am thinking could be naturally expressed as follows: "There is a hole in the ozone layer". If I focus on the wide content, then I can consider not only whether this thought is true in the actual world, where I am thinking it, but also

[32] "Narrow Content", p. 135

> whether it would be true if, say, human beings had never existed to pollute the atmosphere. But we can't evaluate the realization conditions in such a possible world, since the thought won't exist there. We might try to extend the procedure to apply to possible worlds not containing the thought by asking a counterfactual question about the thought: what *would* the content of the thought be if it were thought there? This would work in some cases, but it is likely to leave a large amount of indeterminacy in the narrow content of our thoughts. Consider a version of another one of our stories: Bert, who is a little confused about arthritis, thinks his father has this disease in his thigh. [...] Now consider a possible world in which both the language and the state of medical knowledge are quite different. In this world, there is no word like 'arthritis', in Bert's language, and Bert has no thoughts about his father's health. Are the realization conditions for Bert's actual thought satisfied in this counterfactual world? Bert didn't have the belief there, but if he had, would it have been true? So far as I can see, there is no way to tell.[33]

In language of this paper, this comes to the objection that the first intension of a concept and the notional content of a thought will be indeterminate when evaluated at many centered worlds. There are a number of things to say to this. First, as Stalnaker observes, the truth-value of a thought in a centered world will be most straightforward when the centered world contains the thought at its center, but we have seen that this is not required. To determine the truth-value of a thought in a centered world, we ask: would the thought be true if that world turned out to be actual? Consider my thought "I am an artichoke". If the actual world turned out to be a world with an artichoke at its center, then this thought would be true, even if the artichoke could not think. We can retain the thought from the *real* actual world, and simultaneously ask questions about its truth-value if other worlds turned out to be actual, without any loss in coherence. After all, worlds with an artichoke at their center are precisely the sort of centered worlds that my thought "I am an artichoke" is meant to endorse!

We can handle Stalnaker's second example in a similar way. If Bert's actual world turned out to be a world in which no-one used any terms like "arthritis", then to what would his semantically deferential concept "arthritis" refer? It seems most natural to say that it would refer to nothing at all. This would be a world about which his *background assumptions* would be very mistaken. The use of many concepts depends on certain background assumptions, as Russell's example "The present King of France" shows. If those assumptions do not hold in the actual world, it seems reasonable to say that thoughts involving these concepts lack truth-value. When Bert uses a concept such as "arthritis" with semantic deference, he makes a background assumption that people in the community around me use the word or something like it. If they do not, then all bets are off.

In any case, Bert's thoughts involving "arthritis" still have a significant notional content. They rule in and rule out certain centered worlds in which the background assumptions are

[33] "Narrow Content", pp. 136-7.

satisfied. They take no stand on worlds in which the background assumptions are not satisfied, but that was never their purpose. For the usual explanatory purposes, the inclusion and exclusion of worlds relative to those assumptions will suffice. Centered worlds in which those assumptions fail will usually be ruled out in any case by other beliefs—namely his metalinguistic beliefs in the assumptions in question (for example, his belief that the term "arthritis" is used by the community around him), so that his attitude toward these worlds can be handled in other ways. For the usual explanatory purposes, however, a belief ascriber will be concerned with a thinker's attitude toward worlds that are closer to home, so it is not a problem that notional content and first intensions are only partial functions.

Another class of concepts whose first intensions have a limited domain of definition are perceptual demonstratives. When I think something like "That is pretty", the referent of my demonstrative is often picked out (very roughly) as the cause of such-and-such experience. In a centered world in which a pretty thing is appropriately responsible for the relevant experience at the center, the thought will be true; in a world in which an ugly thing is responsible, the thought is false; but in a world in which there is no appropriate experience at the center, the thought may lack truth-value. In this case fixation of reference relies on the fact that such an experience is present, just as fixation on reference in the "arthritis" case relies on the existence of a surrounding community using the term. Again, it is not a problem that worlds are ruled in and ruled out only relative to a background condition. If for some purpose we wish to invoke the thinker's attitudes to other worlds (a world where the experience is absent, for instance), we can rely on the notional content of other thoughts in the vicinity, such as "The object in front of me is pretty".

The precise domain of definition for the notional contents of particular thoughts is an important issue, for which there will be different answers depending on the thoughts in question. Some concepts, such as pure indexicals ("I", "here", and "now"), have first intensions that are defined across all centered worlds. Pure non-indexicals—concepts whose referent at a centered world does not depend on the position of the center —will also have a wide domain of definition. It is concepts in between, for whom the existence of a referent depends on background assumptions about how things are around the center of a world—as with semantically deferential concepts and certain demonstratives—that will have a restricted domain. This is a deep issue that deserves a detailed treatment of its own, but for now it suffices to note it is no objection to this account to note that the relevant functions are often partial.

Finally, Stalnaker suggests:

> The source of all these problems with this procedure for defining narrow content is the fact that according to this account, narrow content is derivative from (actual and possible) wide content. The explanation of realization conditions takes for granted that somehow, the external and internal facts about a mental event or state that we have identified determine a content (in the ordinary sense) for that state. Then from the actual wide content, together with the facts about what the wide content

> would be under various counterfactual circumstances, we extract the narrow content. Narrow content, as defined by this procedure, presupposes rather than explains wide content.[34]

This, it seems to me, is false and importantly so. It may be true that on Stalnaker's construction, where narrow content is indirectly constructed as the "diagonal" proposition of a two- dimensional function, narrow content is derivative on wide content, but this is an inessential artifact of the construction. As we have defined notional content, it is in no way derivative on relational content. There are two basic notions—what it is for a thought to be true *in* a world and what it is for a thought to be true *of* a world—and neither of these is obviously derivative on the other. Only if we have *already* adopted a framework in which relational content is primary will we determine the notional content of a thought by first determining its relational content in centered world and then evaluating that relational content in the same world. It is in many ways more natural to determine notional content directly, and to directly evaluate the truth of a thought in a world.

Indeed, in many ways relational content seems to be derivative on notional content. We can see this most clearly by looking at concepts, rather than complete thoughts. We have already seen that the second intension of most concepts generally depends on the first intension. In some cases it is a near-copy of the first intension, as for simple descriptive concepts. In more interesting cases, the second intension is determined by evaluating the first intension at the actual world and picking out that value at all possible worlds, as for rigidly designating concepts. In a case like this, the second intension is entirely dependent on how reference is fixed in the actual world, and this is in turn determined by the first intension. By contrast, we can tell the entire story about the first intension without ever involving the second intension. It therefore seems that if either intension is to be primary, it is the first intension. Correspondingly, if either notional content or relational content is to be primary, notional content is the much more natural candidate. Indeed, if we see notional and relational content as composed from first and second intensions respectively, and if we see second intensions as derivative on first intensions, then it seems that narrow content may explain wide content after all.[35]

It seems to me that the shoe is on the other foot, and that Stalnaker's dismissal of notional content is precisely due to his assumption, very common in the recent literature, that relational content is the primary sort of content. In a similar way, Fodor and White were led to their

[34] "Narrow Content", p. 137.

[35] The dependence of second intensions on first intensions, and of relational content on notional content, has implications for the project of giving an explanation of content in naturalistic terms. Arguably, the reason that contemporary causal theories of content have been unsuccessful is that all these attempt to give an account of relational content directly, without addressing notional content first. If the second intension of a concept depends on the first intension, as the current framework suggests, then it is natural to suppose that a naturalistic account of the second intension will require an account of the first intension. Of course I have said nothing here about how a naturalized account of notional content might be given, or indeed about whether such an account is possible (although I believe that it is).

cumbersome characterizations of narrow content by such an assumption. It is unclear why this assumption is so prevalent. One reason is no doubt a concern with language; as we have seen, relational content is often what is communicated in an utterance. Perhaps a second reason is an aversion to indexical accounts of content, grounded in Frege's insistence that the truth-value of a thought cannot depend on a thinker's location in a world. A third reason is that the structure of relational content is easier to express in language than that of notional content: the relational content of a thought such as "London is pretty" is true at all worlds in which London is pretty, but the notional content is more complex, and its discussion is correspondingly more difficult.

Whatever the reasons, the emphasis on relational content has been misplaced. In many ways, notional content is more central to thought than relational content: it is cognitively accessible, it governs the cognitive and rational relations between thoughts, and it governs behavior. But we need not over-react by dismissing relational content entirely. Its utility is still an open question, and it will surely have a central role to play at least in a theory of communication. The most satisfactory overall picture will emerge if we allow the two kinds of content to co-exist alongside each other in a two-dimensional conception of meaning.